

Llama 3.2

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Demo Environment

Development Board: Jetson Orin series motherboard

Note: Due to performance limitations, the Jetson Orin Nano 4GB requires the reduced-parameter version

Meta Llama 3.2 is a series of advanced open-source large-scale language models (LLMs) developed by the Meta AI department.

1. Model Size

Model	Size
llama3.2:1b	1.3GB
llama3.2:3b	2.0GB

2. Performance

Lightweight instruction-tuned benchmarks

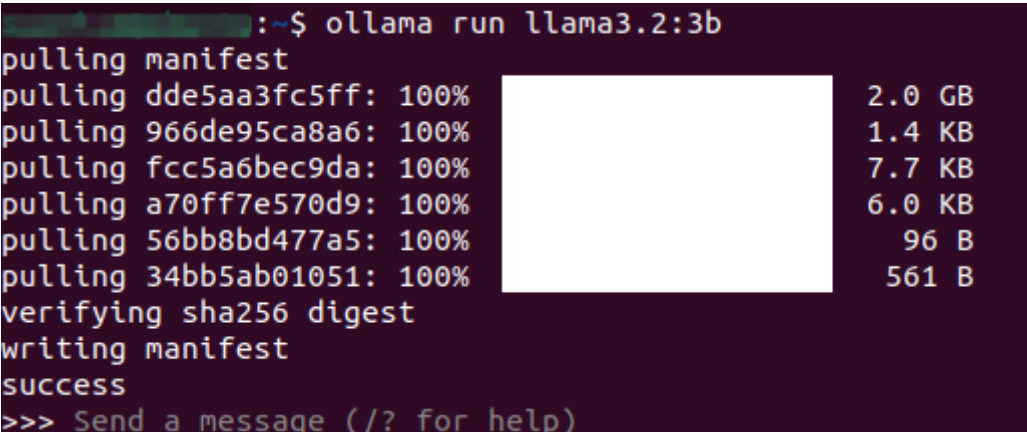
Category Benchmark	Llama 3.2 1B	Llama 3.2 3B	Gemma 2 2B IT (measured)	Phi-3.5-mini IT (measured)
General				
MMLU (5-shot)	49.3	63.4	57.8	69.0
Open-rewrite eval (0-shot, rougeL)	41.6	40.1	31.2	34.5
TLDR9+ (test, 1-shot, rougeL)	16.8	19.0	13.9	12.8
IFEval	59.5	77.4	61.9	59.2
Tool Use				
BFCL V2	25.7	67.0	27.4	58.4
Nexus	13.5	34.3	21.0	26.1
Math				
GSM8K (8-shot, CoT)	44.4	77.7	62.5	86.2
MATH (0-shot, CoT)	30.6	48.0	23.8	44.2
Reasoning				
ARC Challenge (0-shot)	59.4	78.6	76.7	87.4
GPQA (0-shot)	27.2	32.8	27.5	31.9
Hellaswag (0-shot)	41.2	69.8	61.1	81.4
Long Context				
InfiniteBench/En.MC (128k)	38.0	63.3	—	39.2
InfiniteBench/En.QA (128k)	20.3	19.8	—	11.3
NIH/Multi-needle	75.0	84.7	—	52.7
Multilingual				
MGSM (0-shot, CoT)	24.5	58.2	40.2	49.8

3. Using Llama 3.2

3.1 Running Llama 3.2

Use the run command to run the model. If the model is not already downloaded, it will automatically pull the model from the Olama model library:

```
ollama run llama3.2:3b
```



3.2 Starting a Conversation

How many minutes are there in a day?

Response time depends on your hardware configuration, so please be patient!

```
>>> How many minutes are there in a day?
There are 1440 minutes in a day. This is calculated by multiplying the
number of hours in a day (24) by the number of minutes in an hour (60).

24 hours/day * 60 minutes/hour = 1440 minutes/day

>>> Send a message (/? for help)
```

3.3 Ending a Conversation

Use the `Ctrl+d` shortcut or `/bye` to end a conversation!

References

Ollama

Official Website: <https://ollama.com/>

GitHub: <https://github.com/ollama/ollama>

Llama 3.2

Official Website: https://www.llama.com/docs/model-cards-and-prompt-formats/llama3_2/

Ollama Model: <https://ollama.com/library/llama3.2>